

Mutation of Local Floer Chain Complexes across Hamiltonian Bifurcations

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Contents

Based on a joint work with Hong-Kwon Jo (SNU), arXiv:2607.22178.

Main result. We describe the explicit changes of the local Floer chain complex for generic types of Hamiltonian bifurcations.

- **Motivation and Preliminaries.**

Bifurcations, the Conley–Zehnder index, and Floer homology.

- **Methodology.**

Model Hamiltonian, Morse–Bott type cascade counting.

- **Chain complex mutation.**

Result for 6 types of bifurcations.

(Generic 4 types, $\mathbb{Z}_2$ -symmetric 2 types.)

Motivation and Preliminaries

Motivation 1. Bifurcations and Conley–Zehnder indices

A **bifurcation** is, heuristically, the birth or death of periodic orbits.

Several recent works [AFvK⁺23], [AB25], [JKvK26] relate bifurcations to symplectic invariants, for example Conley–Zehnder indices.

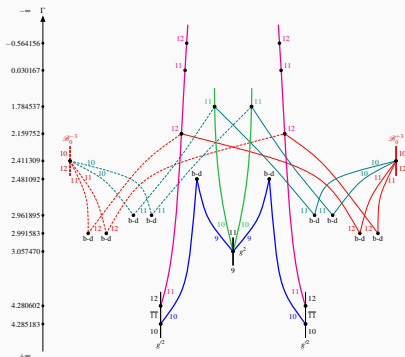


Figure 1: Bifurcation graph of Hill's lunar problem, [AB25].

Can we make a *symplectic dictionary* for bifurcation types?

Motivation 2. Rotating Kepler Problem

In the rotating Kepler problem, there are three important periodic orbits: retrograde, direct and vertical collision orbits.

The multiple covers of the retrograde and direct orbits bifurcate, producing orbits that look like flowers.

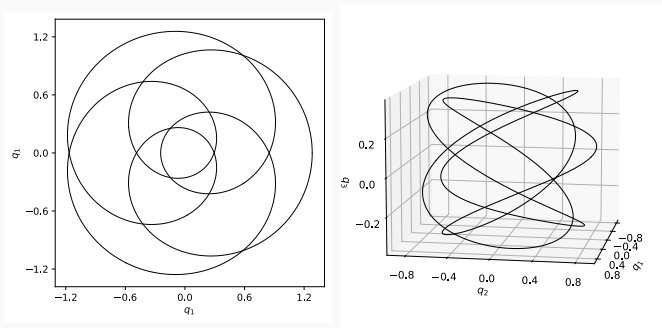
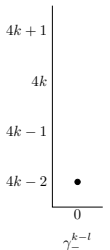
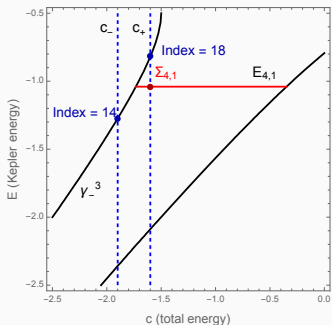


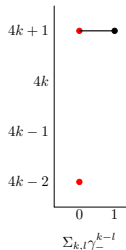
Figure 2: Resonant orbits of RKP, courtesy of Chankyu Joung.

Motivation 2. Rotating Kepler Problem

In my previous work [Lee26], I observed that this bifurcation can be translated into a change in the Floer chain complex.



At $c_{k,l}^- - \epsilon$



At $c_{k,l}^- + \epsilon$

Slogan. A bifurcation gives a *mutation of a Floer chain complex*.

We aim to describe this effect explicitly.

Preliminaries

Hamiltonian System and Periodic Orbits

Let (W, ω) be a symplectic manifold and $H : W \rightarrow \mathbb{R}$ be a function.

We define a **Hamiltonian vector field** by a relation $i_{X_H} \omega = -dH$.

Note. If $W = T^*\mathbb{R}^n$, $\omega = dp \wedge dq$ and $H = |p|^2/2 + V(q)$, X_H gives usual Hamiltonian equation in the classical mechanics,

$$\dot{q} = p, \quad \dot{p} = -V'(q).$$

Main interest is **periodic orbits**, which is an orbit γ such that $\dot{\gamma} = X_H$ and $\gamma(0) = \gamma(T)$ for some $T > 0$.

Periodic orbits form backbone of the dynamics, and also connects the topology and dynamics. (Arnold conjecture, Floer theory)

Poincaré Section

A **Poincaré section** is a transversal section $\Sigma \subset H^{-1}(c)$ at $p \in \gamma$.

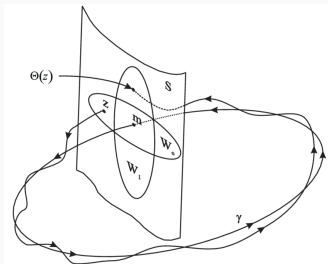


Figure 3: Poincaré section [OR99].

A **return map** is defined by $\Phi : U \subset \Sigma \rightarrow \Sigma$ by

$$\Phi(q) = Fl_{\tau(q)}^{X_H}(q), \quad \tau(q) = \min \left\{ t : Fl_t^{X_H}(q) \in \Sigma \right\}.$$

Note. The return map of Hamiltonian system is a symplectomorphism.

Floquet multipliers are eigenvalues of $d_p \Phi$.

Stability

If $\dim W = 4$, so $\dim \Sigma = 2$, then $\lambda \in S^1 \cup \mathbb{R} \subset \mathbb{C}$.

We call an orbit is **elliptic** if $\lambda \in S^1$ and **hyperbolic** if $\lambda \in \mathbb{R} \setminus \{0\}$.

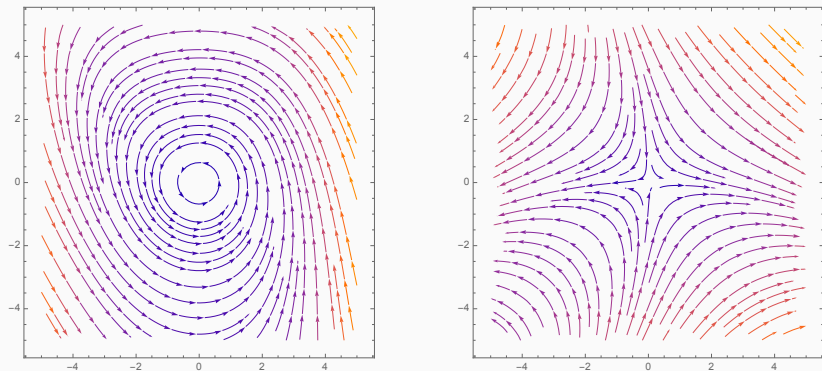


Figure 4: Elliptic and hyperbolic fixed points.

Bifurcation

Let $H_\varepsilon : W^4 \rightarrow \mathbb{R}$ be a 1-parameter family of Hamiltonians, and let γ_0 be a periodic orbit of H_0 with Floquet multiplier λ_0 .

If $\lambda_0 \neq 1$, so the orbit is **non-degenerate**, then nearby periodic orbits form an *orbit cylinder*

$$\Gamma = \{(\gamma_\varepsilon, \varepsilon) : \gamma_\varepsilon \text{ is a periodic orbit of } H_\varepsilon\}.$$

This is a *continuation* of the orbit along the bifurcation parameter.

We follow γ_ε and detect λ_ε . If $\lambda_\varepsilon = 1$ at some ε , then γ_ε is degenerate, and the orbit cylinder may break:

- periodic orbits may be born or vanish; this is called a **bifurcation**,
- if $\lambda_\varepsilon \neq 1$ but $\lambda_\varepsilon^k = 1$, then the k -th cover bifurcates.

Example. Period Doubling

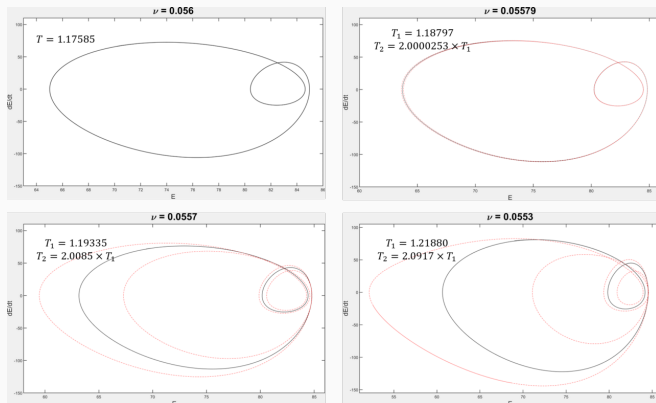


Figure 5: Period doubling in Kuramoto–Sivashinsky equation [Max21].

Period doubling is a generic bifurcation of a double cover.

The red dashed orbit is the new orbit that emerges from the bifurcation.

Conley–Zehnder Index

The **Conley–Zehnder index** $\mu_{CZ}(\gamma)$ is an integer associated with a periodic orbit and can be understood as a *symplectic winding number*.

We are interested in a change of index along the parameter; $\mu_{CZ}(\gamma_\epsilon)$.

Along a non-degenerate family, the CZ index is constant.

Key idea. A change in μ_{CZ} = Degeneracy of γ_ϵ = A bifurcation occurs

Hence, a bifurcation can be detected via change of the CZ index.

Note. This is also useful in practical purposes. ([AB25], [JKvK26].)

Floer Homology

Floer homology $HF_*(W, H)$ is a topological invariant associated with a symplectic manifold W and Hamiltonian H . (cf. [AD14].)

This is determined by the **Floer chain complex** $CF_*(W, H)$, whose

- generators are 1-periodic orbits of H ,
- grading is given by CZ indices,
- the differential is determined by counting **Floer cylinder**, whose boundaries are asymptotic to periodic orbits.

We will use **local Floer homology**, which is a Floer homology on a neighborhood of a periodic orbit, $HF_*^{\text{loc}}(\gamma_0)$.

Across a bifurcation, CF_* changes substantially;

- orbits are born or vanish $\Rightarrow$ the number of generators changes.
- the CZ index changes $\Rightarrow$ grading changes

However, the homology doesn't change, so the differential is responsible for keeping the homology invariant. We call this **mutation** of LFCC.

Methodology

We're interested in the case such that $\lambda^k = 1$.

1. By **Birkhoff normal form**, find a generating Hamiltonian G_ε of Φ^k .
2. Construct **Model Hamiltonian** H_ε from G_ε .
3. Using **Floer cascade** to compute Floer differential.

Birkhoff Normal Form

We find the fixed points of the iterated return map Φ_ε^k .

This can also be done in the essentially same way by finding critical points of the generating Hamiltonian G_ε of Φ_ε^k .

1. If $\lambda = \exp(i\alpha)$, we can find a formal power series $G = \sum G_n$ of the generating Hamiltonian of Φ^k of the form

$$G_t = G_2 + \sum_{n \geq 3} G_{n,t}$$

where $\deg G_n = n$, $G_2^{ss} = \alpha(u^2 + v^2)$, $\dot{G}_{n,t} = \{G_2^{ss}, G_{n,t}\}$.

2. If $\lambda^k = 1$, $G_2^{ss} = 2\pi l(u^2 + v^2)$, which means that G is generated by polynomials commuting with rotation by $2\pi l$, i.e.,

$$G \in \mathbb{R}[\rho^2, \rho^k \cos(k\varphi - 2\pi lt), \rho^k \sin(k\varphi - 2\pi lt)].$$

3. Using rotating frame, we remove the time-dependency and G becomes an **autonomous $\mathbb{Z}_k$ -symmetric polynomials**.

Birkhoff Normal Form

Ring of $\mathbb{Z}_k$ -symmetric polynomials is $\mathbb{R}[\rho^2, \rho^k \cos k\varphi, \rho^k \sin k\varphi]$.

$\Rightarrow$ We can write the Hamiltonians by

$$G_\varepsilon = \begin{cases} \alpha u^2 + \beta v^3 + \varepsilon(au + bv + \dots) & k = 1, \\ \alpha u^2 + \beta v^4 + \varepsilon(av^2 + \dots) & k = 2, \\ \alpha \rho^4 + \beta \rho^k \cos k\varphi + \varepsilon(a\rho^2 + \dots) & k \geq 3. \end{cases}$$

Note. Only leading terms are relevant to observe critical points.

The critical points of G are fixed points of Φ^k , which is k -periodic points of Φ , so there is a **k -to-1 correspondence** except for the origin.

Model Hamiltonian

Let $\tilde{H}_\varepsilon$ be a Hamiltonian and G_ε be a generating Hamiltonian of Φ_ε^k .

We construct a model Hamiltonian on $I_r \times (\mathbb{R}/k\mathbb{Z})_\theta \times \Sigma_z$ by,

$$H_\varepsilon = kh(r) + G_\varepsilon(z).$$

where $h'(0) = 1$ and $h''(0) \neq 0$.

- k -to-1 correspondence between $\text{Crit}(G_\varepsilon)$ and $\text{Per}(H_\varepsilon)$, say $p \leftrightarrow \gamma_p$.
- $\mu_{CZ}(\gamma_p) = 1 - \text{Ind}_{\text{Morse}}(p)$.
- A gradient flow between p and q gives a Floer cylinder between γ_p and γ_q , which we call a **gradient revolution**.

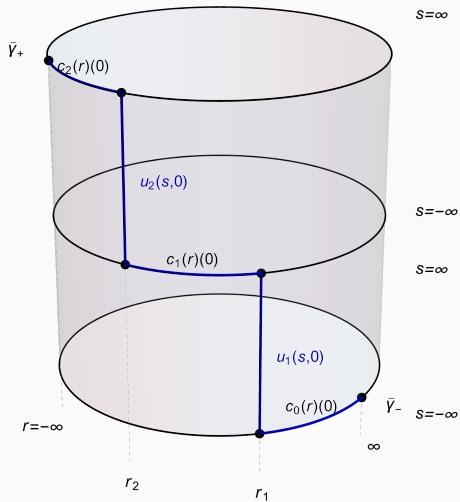
Note. H_ε and $\tilde{H}_\varepsilon$ are different, but has the same return map, bifurcation type, and Floer-type data up to finite degree and symplectomorphism.

Our periodic orbits have an S^1 -symmetry under time shifts, which makes the situation slightly different from the usual Floer theory.

We equip each orbit with a Morse function, and consider two critical points ($\max \hat{\gamma}$ and $\min \check{\gamma}$) as generators of a Floer chain complex.

In this case, the differential is given by counting **Floer cascades**; a sequence of Morse trajectories and Floer cylinders.

Floor Cascade



Floer Cascade

If there is a Floer cylinder $u(s, t)$ from γ_+ to γ_- , for $p \in \gamma_{\mp}$, we define

$$\text{ev}^{\pm}(p) = \left\{ q \in \gamma_{\pm} : \text{for } t_0 \text{ such that } p = \lim_{s \rightarrow \mp \infty} u(s, t), q = \lim_{s \rightarrow \pm \infty} u(s, t) \right\}.$$

Then we count

1. $(\mu(\gamma_+; \gamma_-) = 1)$ Morse trajectory on γ_- from $\hat{\gamma}_+$ to $\text{ev}^-(\hat{\gamma}_-)$, or one on γ_+ from $\text{ev}^+(\check{\gamma}_+)$ to $\check{\gamma}_-$.
2. $(\mu(\gamma_+; \gamma_-) = 2)$ Morse trajectory on γ_0 (such that $\mu(\gamma_0; \gamma_-) = 1$) from $\text{ev}^+(\check{\gamma}_-)$ to $\text{ev}^-(\hat{\gamma}_+)$.

Floor Cascade

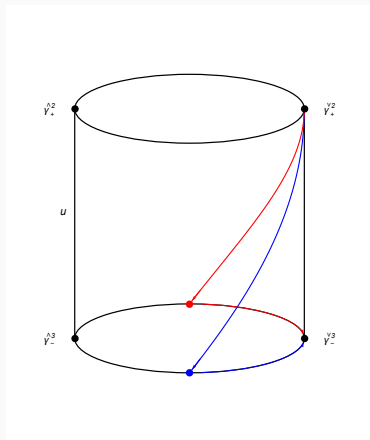
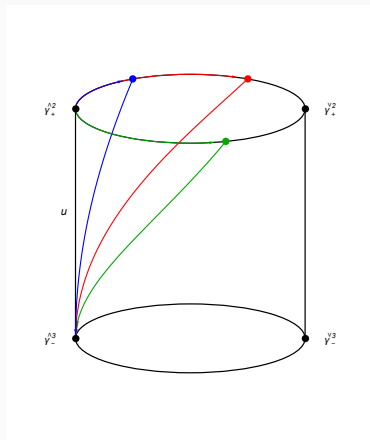


Figure 6: Cascades between 2nd and 3rd covers.

Cascade Counting Theorem

Theorem

Let $\gamma_{\pm}$ be two non-constant $1/k_{\pm}$ -periodic orbits, so that $\gamma_{\pm}^{k_{\pm}}$ are 1-periodic orbits. Moreover, assume that

$$\mu_{CZ}(\gamma_+^{k_+}) - \mu_{CZ}(\gamma_-^{k_-}) = 1.$$

Then, the quotient $\mathcal{M}_0(\gamma_-^{k_-}, \gamma_+^{k_+})/S^1$ is a 0-dimensional $\mathbb{Z}_{(k_+, k_-)}$ -orbifold, and we have

$$\begin{aligned}\#_2 \mathcal{M}(\hat{\gamma}_-^{k_-}, \hat{\gamma}_+^{k_+}) &\equiv k_- \cdot \#(\mathcal{M}_0(\gamma_-^{k_-}, \gamma_+^{k_+})/S^1) \pmod{2}, \\ \#_2 \mathcal{M}(\check{\gamma}_-^{k_-}, \check{\gamma}_+^{k_+}) &\equiv k_+ \cdot \#(\mathcal{M}_0(\gamma_-^{k_-}, \gamma_+^{k_+})/S^1) \pmod{2}.\end{aligned}$$

In particular, if there is only one cylinder from a simple γ_0 to a multiple cover γ_1^k ,

$$\partial \hat{\gamma}_0 = k \hat{\gamma}_1^k, \quad \partial \check{\gamma}_0 = \check{\gamma}_1^k.$$

Chain Complex Mutation

Meyer's Classification

Generic Hamiltonian bifurcations are classified in [Mey70].

They are organized by the covering number k of the underlying orbit.

- $k = 1$: birth-death
- $k = 2$: period doubling, materialization (opposite stability types)
- $k = 3, 4$: phantom kiss
- $k \geq 4$: emission

Aside. We will use names from [AM78], where three extra types involving critical points are given.

Other names: touch-and-go, period tripling/quadrupling, etc.

Note. The fourth cover is a borderline case, depending on the coefficients of the normal form.

Birth-death ($k = 1$)

A generic single cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(u, v) = \alpha v^2 + \beta u^3 + \dots + \varepsilon(u + \dots).$$

- ($\varepsilon < 0$) There are no periodic orbits.
- ($\varepsilon > 0$) One elliptic γ_0 and one hyperbolic orbit γ_1 are born.
- There is one gradient trajectory between them.

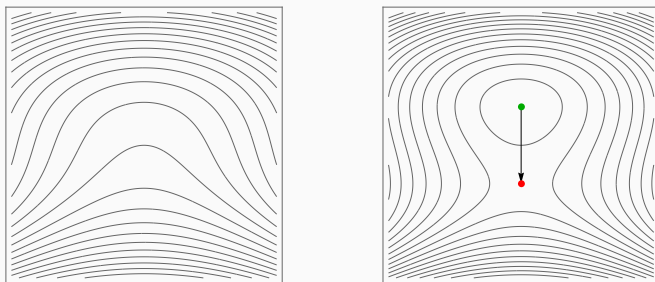


Figure 7: Critical points and gradient trajectories of birth-death bifurcation
In every picture, ● is minimum, ● is saddle, ● is maximum.

Birth-death ($k = 1$)

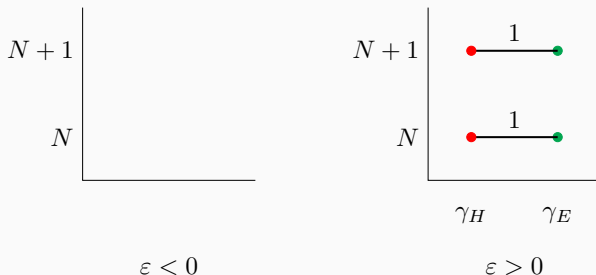


Figure 8: Chain complex mutation across the birth-death.

- Points at (n, m) corresponds to a generator of degree $n + m$.
- The line indicates the differential; $\partial(\hat{\gamma}_E) = \hat{\gamma}_H$ here.
- We are using $\mathbb{Z}_2$ -coefficients.

We have $HF_*^{\text{loc}}(\gamma) = 0$, where γ is the degenerate orbit at $\varepsilon = 0$.

Period doubling ($k = 2$)

A generic double cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(u, v) = \alpha v^2 + \beta u^4 + \dots + \varepsilon(u^2 + \dots).$$

- ($\varepsilon < 0$) One elliptic orbit γ_0^2 exists.
- ($\varepsilon > 0$) γ_0^2 becomes hyperbolic, and an elliptic orbit (●) γ_{new} is born.
- There is one cylinder between γ_0^2 and γ_{new} .

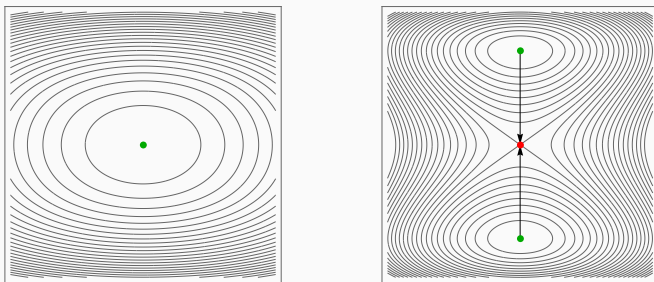


Figure 9: Period doubling bifurcation

Period doubling ($k = 2$)

- $\mu_{CZ}(\gamma_+^2; \gamma_-^2) = -1$ or 1 . We assume -1 below.
- $\mu_{CZ}(\gamma_{new}; \gamma_-^2) = 0$.

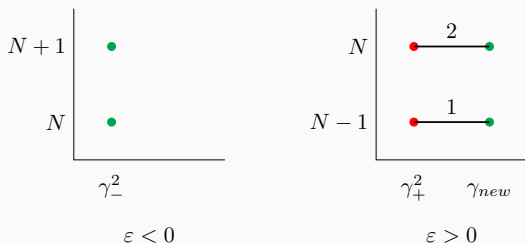


Figure 10: Chain complex mutation across the period doubling.

$\check{\gamma}_H$ and $\check{\gamma}^2$ cancel each other, while $\hat{\gamma}_H$ and $\hat{\gamma}^2$ do not.

$$HF_*^{\text{loc}}(\gamma_0^2) \cong \begin{cases} \mathbb{Z}_2 & \text{if } * = 0, 1 \text{ relative to } \gamma_-^2, \\ 0 & \text{otherwise} \end{cases}$$

Phantom kiss ($k = 3, 4$)

A generic 3rd or 4th cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(\rho, \varphi) = \alpha \rho^k \cos k\varphi + \dots + \varepsilon(\rho^2 + \dots).$$

- ($\varepsilon < 0$) One elliptic orbit γ^k and one hyperbolic orbit γ_H .
- ($\varepsilon = 0$) They meet, and suddenly become unstable.
- ($\varepsilon > 0$) The same configuration, but the index of γ^k changes.
- One cylinder exists between γ^k and γ_H .

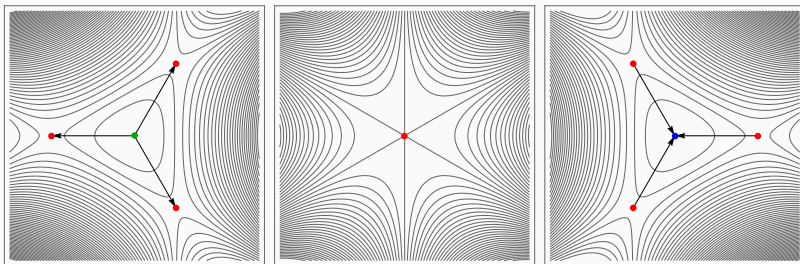


Figure 11: Phantom kiss bifurcation

Phantom kiss ($k = 3, 4$)

1. $\mu_{CZ}(\gamma_+^k; \gamma_-^k) = \pm 2$. Here, we assume -2 .
2. $\mu_{CZ}(\gamma_{H,\pm}; \gamma_-^k) = \pm 1$, say -1 , and it doesn't change.

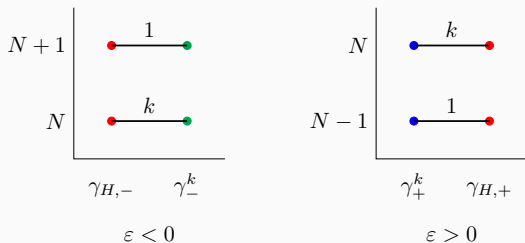


Figure 12: Chain complex mutations across the phantom kiss

$$HF_*^{\text{loc}}(\gamma_0^3) = 0, \quad HF_*^{\text{loc}}(\gamma_0^4) \cong \begin{cases} \mathbb{Z}_2 & \text{if } * = -1, 0 \text{ relative to } \gamma_-^4, \\ 0 & \text{otherwise} \end{cases}$$

Emission ($k \geq 4$)

Generic $k \geq 4$ -th cover bifurcation, with generating Hamiltonian

$$G_\varepsilon(\rho, \varphi) = \alpha\rho^4 + \beta\rho^k \cos k\varphi + \dots + \varepsilon(\rho^2 + \dots).$$

- ($\varepsilon < 0$) One elliptic orbit γ_-^k exists.
- ($\varepsilon > 0$) γ_-^k is still elliptic, and a new elliptic orbit γ_E and a new hyperbolic orbit γ_H are born.
- Two cylinders from γ_E to γ_H , and one cylinder from γ_H to γ_+^k .

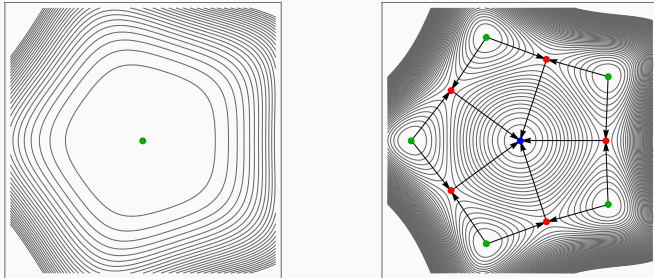


Figure 13: Emission bifurcation

Emission ($k \geq 4$)

1. $\mu_{CZ}(\gamma_+^k; \gamma_-^k) = \pm 2$, say -2 .
2. $\mu_{CZ}(\gamma_H; \gamma_-^k) = -1$, $\mu_{CZ}(\gamma_E; \gamma_-^k) = 0$.

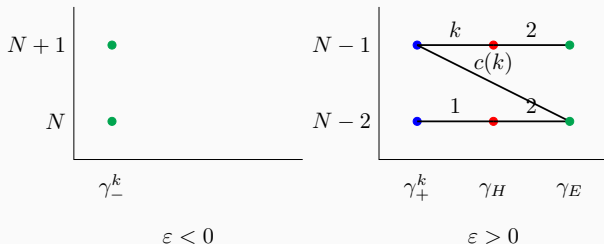


Figure 14: Chain complex mutations across the emission

$c(k)$ depends on the choice of Morse function, but always odd if k is even.

$$HF_*^{\text{loc}}(\gamma_0^k) \cong \begin{cases} \mathbb{Z}_2 & \text{if } * = 0, 1 \text{ relative to } \gamma_-^k, \\ 0 & \text{otherwise} \end{cases}$$

Emission ($k \geq 4$)

To find $c(k)$, we need to count the cascade of length 2 from $\check{\gamma}^k$ to $\hat{\gamma}_E$, which is equivalent to a count of Morse trajectories

$$\text{ev}^-(\check{\gamma}_E) \rightarrow \text{ev}^+(\hat{\gamma}^k) \text{ on } \gamma_H.$$

This depends on the choice of the Morse function on γ_H .

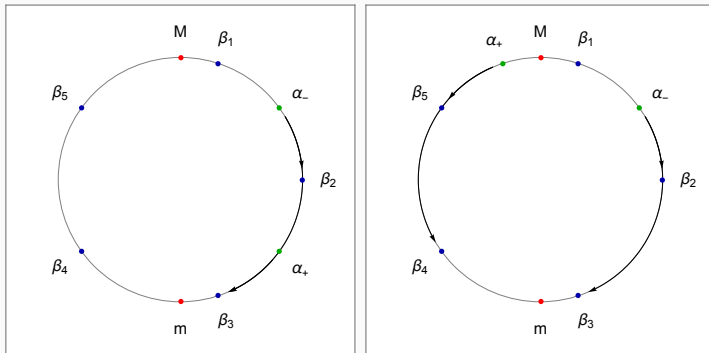


Figure 15: Some cases of $c(k)$.

Pitchfork ($\mathbb{Z}_2$ -symmetric, $k = 1$)

For $\mathbb{Z}_2$ -symmetric systems, we impose an extra $\mathbb{Z}_2$ -symmetry on G_ε .

$\Rightarrow$ We analyze $\mathbb{Z}_{\text{lcm}(2,k)}$ -symmetric polynomials, so only odd-cover bifurcation produces new phenomena.

For $k = 1$, we have period doubling picture, where now two new critical points correspond to **two** new periodic orbits.

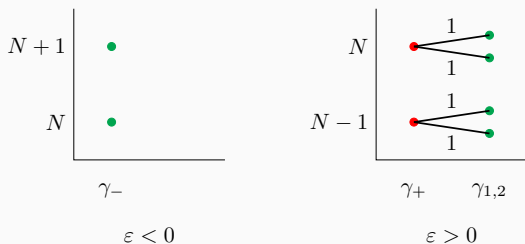


Figure 16: Chain complex mutations across the pitchfork

Double emission ($\mathbb{Z}_2$ -symmetric, $k \geq 3$ odd)

For $k \geq 3$, we have $2k$ -emission.

- There are two new hyperbolic and two new elliptic orbits.
- Cylinder configuration is changed. (One cylinder from each $\gamma_{E,i}$ to $\gamma_{H,j}$, and one cylinder from each $\gamma_{H,j}$ to γ^k .)

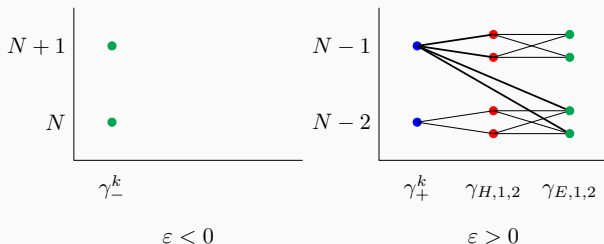


Figure 17: Chain complex mutations across the double emission

Further discussions

- Orientation and $\mathbb{Z}$ -coefficients.
- More symmetries, multi-parameter, higher dimensional bifurcations.
- This gives a *generic* change of local Floer chain complex.
⇒ Persistent module description and Floer-type Cerf theory?
- Bifurcation of symmetric orbits can be described as a Lagrangian Floer theory ⇒ Similar work for Lagrangians?



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Illustration of period doubling in the Kuramoto–Sivashinsky equation. CC0 1.0 Universal Public Domain Dedication.



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Figure 18: Thank you for your attention!

Happy birthday Professor Cieliebak!